

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

REPORT ON TWO-DAY WORKSHOP

“Future Forward: Empowering Minds for Placements”

The Department of Computer Science and Engineering at ATME College of Engineering, Mysuru, organized a comprehensive **two-day workshop titled “Future Forward: Empowering Minds for Placements”** on 10th and 11th November 2025. The workshop aimed to prepare final-year CSE students to meet the expectations of the IT industry by strengthening their technical foundations and problem-solving skills.

The modern IT job market demands strong conceptual clarity, in-depth technical knowledge, and practical application skills. With placement drives approaching, this workshop was organized to enhance student readiness through well-structured sessions on **Java Programming, Object-Oriented Programming (OOPs), Data Structures, and Algorithms**. The sessions were conducted by experienced faculty members of the department—**Dr. Shilpa B. L.** and **Mr. Teja N.**—who brought in valuable insights, real-time examples, industry expectations, and hands-on demonstrations.

Objectives of the Workshop

The workshop was designed with the following key objectives:

1. Strengthen technical fundamentals essential for placement-oriented assessments.
2. Provide a clear understanding of Java programming concepts and best practices.
3. Reinforce Object-Oriented Programming principles with real-world examples.
4. Enhance student competency in Data Structures and Algorithms, crucial for coding rounds.
5. Develop the ability to approach problem-solving logically and efficiently.
6. Improve placement preparedness by offering exposure to industry-style coding patterns.

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7. Build confidence among students by encouraging active participation and hands-on coding practice.

Detailed Description – Day 1

The first day of the workshop focused on creating a strong foundation in **Java programming**. Mr. Teja N. began the morning session by explaining why Java remains one of the most demanded programming languages in placement interviews. Students learned about JVM architecture, Java's memory model, and writing simple programs.

In the second session, students practiced **loops, conditional statements, and function design**, enabling them to write modular and readable code. Several live coding examples were demonstrated, which helped students clearly understand program flow.

Post-lunch, **OOPs** concepts were introduced by Dr. Shilpa B. L. Students explored real-world analogies for OOPs pillars:

- **Abstraction** using ATM machine examples
- **Encapsulation** using class-based access control
- **Inheritance** using academic hierarchy
- **Polymorphism** through method overloading and overriding

The final session emphasized **interfaces, abstract classes, and code reusability**, preparing students for advanced problem-solving.

Session Time	Topic	Resource Person	Highlights
9:00 – 10:30 AM	Introduction to Java Programming	Mr. Teja N.	Basics of Java, JDK/JRE, syntax, variables, data types
10:45 – 12:30 PM	Control Structures & Functions	Mr. Teja N.	Loops, conditionals, methods, modular programming
1:30 – 3:00 PM	Object-Oriented Programming Concepts	Dr. Shilpa B. L.	Classes, objects, abstraction, encapsulation
3:15 – 5:00 PM	Inheritance, Polymorphism, Interfaces	Dr. Shilpa B. L.	Real-world examples, UML mapping, coding demos

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Table 1: Day 1 Session Breakdown

Day 2 – 11th November 2025

Topics Covered: Data Structures & Algorithms

Time: 9:00 AM – 5:00 PM

Detailed Description – Day 2

Day 2 shifted focus toward one of the most crucial components of technical interviews **Data Structures and Algorithms (DSA)**. The morning session introduced foundational structures such as **arrays, linked lists, stacks, and queues**, along with their real-world applications.

The second session concentrated on **trees, binary search trees, and graphs**, providing clarity on hierarchical and network-based data organization. Students learned traversal methods like in-order, pre-order, BFS, and DFS.

In the afternoon, Dr. Shilpa B. L. conducted an in-depth session on **algorithm analysis**, emphasizing Big-O notation, time/space complexity, and optimization strategies. Students practiced implementing common algorithms including insertion sort, binary search, and hashing concepts.

Session Time	Topic	Resource Person	Highlights
9:00 – 10:30 AM	Introduction to Data Structures	Mr. Teja N.	Arrays, linked lists, stacks, queues
10:45 – 12:30 PM	Trees and Graphs	Mr. Teja N.	Traversals, shortest path concepts
1:30 – 3:00 PM	Algorithms & Time Complexity	Dr. Shilpa B. L.	Big-O notation, sorting/searching algorithms
3:15 – 5:00 PM	Placement-Oriented Coding Problems	Dr. Shilpa B. L.	Hands-on coding, live problem-solving

Table 2: Day 2 Session Breakdown

The final session was a **hands-on competitive coding practice**. Students solved placement-style problems using Java and analysed their solutions for efficiency. This session boosted their confidence to tackle coding rounds effectively.

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Student Participation and Engagement

The workshop recorded **high participation** from final-year students. Many students actively engaged in discussions, asked questions, and participated in live coding exercises. The interactive teaching style and hands-on demonstrations helped keep the sessions energetic and student-friendly.

Students appreciated:

- Real-world examples
- Step-by-step coding explanations
- Placement-oriented problem discussions
- Personalized guidance from resource persons.

The two-day workshop “**Future Forward: Empowering Minds for Placements**” successfully achieved its goal of strengthening the technical foundation of final-year CSE students. The focused sessions on **Java, OOPs, Data Structures, and Algorithms** ensured that participants gained practical knowledge directly relevant to placement exams and technical interviews.

The department extends gratitude to the resource persons—**Dr. Shilpa B. L.** and **Mr. Teja N.**—for delivering comprehensive and impactful sessions. The workshop contributed significantly to enhancing students’ confidence and preparedness for upcoming placement drives.

Overall, the workshop provided an excellent platform for students to understand the real-world aspects of software development and strengthened their enthusiasm toward learning and innovation in web and app technologies.

Learning Outcomes

- At the end of the two-day workshop, students gained the following outcomes:
- Clear understanding of Java programming fundamentals.
- Strong grasp of OOPs concepts, essential for technical interviews.
- Ability to implement and visualize core data structures.
- Improved confidence in solving algorithmic problems.
- Better understanding of time complexity and code optimization.
- Enhanced logical reasoning and problem-solving skills.
- Exposure to coding test patterns used by top IT companies.



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PO Addressed	PSO Addressed
PO1, PO2, PO3, PO5, PO9, PO11	PSO1, PSO2



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